ADDING A LARGE TWIST OF A RELATIVELY AMPLE SHEAF GIVES A VERY AMPLE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Very ampleness can be checked after restricting to an affine covering of the base. Lemma \ref{lemma-characterize-very-ample-on-finite-type}, cited at proof lines 9652-9653. Reduces to an affine base.
- The affine characterisation of ampleness and very ampleness of powers. Lemma \ref{lemma-finite-type-over-affine-ample-very-ample} and Properties, Proposition \ref{properties-proposition-characterize-ample}, cited at proof lines 9654-9656 and 9668-9670. Make a twist globally generated and give the immersion.

Source reference. Original lemma 0FVC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19807–19807. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3489–3489. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01VT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2458–2458. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01Q3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2252–2252. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01WD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2478–2478. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07RK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9870–9870. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02V0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3751–3751. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{N}$, $\mathcal{L}$ be invertible $\mathcal{O}_X$-modules. Assume S is quasi-compact, f is of finite type, and $\mathcal{L}$ is f-ample. Then $\mathcal{N} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes d}$ is f-very ample for all $d \gg 1$.*

Proof. By Lemma 02NQ we reduce to the case S is affine. Combining Lemma 01VT and Properties, Proposition 01Q3 we can find an integer d_0 such that $\mathcal{N} \otimes \mathcal{L}^{\otimes d_0}$ is globally generated. Choose global sections $s_0, \ldots, s_n$ of $\mathcal{N} \otimes \mathcal{L}^{\otimes d_0}$ which generate it. This determines a morphism $j : X \to \mathbf{P}^n_S$ over S. By Lemma 01VT we can also pick an integer d_1 such that for all $d \geq d_1$ there exist sections $t_{d,0}, \ldots, t_{d,n(d)}$ of $\mathcal{L}^{\otimes d}$ which generate it and define an immersion

$$j_d = \varphi_{\mathcal{L}^{\otimes d}, t_{d,0}, \ldots, t_{d,n(d)}} : X \longrightarrow \mathbf{P}^{n(d)}_S$$

over S. Then for $d \geq d_0 + d_1$ we can consider the morphism

$$\varphi_{\mathcal{N} \otimes \mathcal{L}^{\otimes d}, s_j \otimes t_{d-d_0,i}} : X \longrightarrow \mathbf{P}^{(n+1)(n(d-d_0)+1)-1}_S$$

This morphism is an immersion as it is the composition

$$X \to \mathbf{P}^n_S \times_S \mathbf{P}^{n(d-d_0)}_S \to \mathbf{P}^{(n+1)(n(d-d_0)+1)-1}_S$$

where the first morphism is (j, j_{d-d_0}) and the second is the Segre embedding (Constructions, Lemma 01WD). Since j is an immersion, so is (j, j_{d-d_0}) (apply Lemma 07RK). We have a composition of immersions and hence an immersion (Schemes, Lemma 02V0). □